



Areas Between Curves Using Sub-Regions

Calculus Worksheet · Grade 11-12

Name: _____

Date: _____

Score: / 10

Learning Objectives

- Identify sub-regions formed by intersecting curves and lines
- Set up definite integrals using the top-minus-bottom function method
- Evaluate definite integrals to compute total area between curves

For each problem, sketch the region, identify sub-regions if needed, set up the appropriate definite integral(s), and evaluate to find the area.

1. Find the area of the region in the first quadrant bounded above by $y = \sqrt{x}$, below by the x-axis, and by the line $y = x - 2$.

$$A = \int_0^2 \sqrt{x} dx + \int_2^4 (\sqrt{x} - (x - 2)) dx$$

Answer: _____

2. Find the area of the region bounded by $y = x^2$ and $y = x$ between $x = 0$ and $x = 1$.

$$A = \int_0^1 (x - x^2) dx$$

Answer: _____

3. Find the area of the region bounded above by $y = 4 - x^2$ and below by the x-axis.

$$A = \int_{-2}^2 (4 - x^2) dx$$

Answer: _____

4. Find the area of the region bounded by $y = \sqrt{x}$, the y-axis, and the line $y = 2$.

$$A = \int_0^2 y^2 dy$$

Answer: _____

5. Find the area of the region bounded by $y = x^3$ and $y = x$ in the first quadrant.

$$A = \int_0^1 (x - x^3) dx$$

Answer: _____



6. Find the area of the region bounded by $y = x + 2$, $y = x^2$, and the y-axis using sub-regions if needed.

$$A = \int_0^2 ((x + 2) - x^2) dx$$

Answer: _____

7. Find the area of the region bounded by $y = \sqrt{x}$, $y = x - 2$, and the line $x = 0$ using sub-regions.

$$A = \int_0^2 \sqrt{x} dx + \int_2^4 (\sqrt{x} - (x - 2)) dx$$

Answer: _____

8. Find the area enclosed between $y = \sin(x)$ and the x-axis from $x = 0$ to $x = \pi$.

$$A = \int_0^\pi \sin(x) dx$$

Answer: _____

9. Find the area of the region bounded by $y = x^2$ and $y = 2x$ using top-minus-bottom.

$$A = \int_0^2 (2x - x^2) dx$$

Answer: _____

10. Find the area of the region bounded by $y = \sqrt{x}$, $y = 6 - x$, and the x-axis using sub-regions.

$$A = \int_0^4 \sqrt{x} dx + \int_4^6 (6 - x) dx$$

Answer: _____





Emphasize that students must sketch graphs first and check whether the bottom (or top) function changes across the interval — this determines whether sub-regions are required.

Solutions

1. Find the area of the region in the first quadrant bounded above by $y = \sqrt{x}$, below by the x-axis, and by the line $y = x - 2$.

$$A = \int_0^2 \sqrt{x} dx + \int_2^4 (\sqrt{x} - (x - 2)) dx$$

- Sketch $y = \sqrt{x}$, $y = x - 2$, and the x-axis to visualize two sub-regions.
- Find the intersection of $y = \sqrt{x}$ and $y = x - 2$ by solving $\sqrt{x} = x - 2$, giving $x = 4$.
- For the first sub-region from $x = 0$ to $x = 2$, integrate $\sqrt{x}$ with the x-axis as the bottom.
- For the second sub-region from $x = 2$ to $x = 4$, integrate $\sqrt{x}$ minus $(x - 2)$.
- Evaluate each integral and add the results to get the total area of $10/3$.

Answer: $\frac{10}{3}$

2. Find the area of the region bounded by $y = x^2$ and $y = x$ between $x = 0$ and $x = 1$.

$$A = \int_0^1 (x - x^2) dx$$

- Sketch both curves and note $y = x$ is above $y = x^2$ on $[0,1]$.
- Set up the integral of top minus bottom: $x - x^2$.
- Integrate to obtain $x^2/2 - x^3/3$.
- Evaluate from 0 to 1 to get $1/2 - 1/3 = 1/6$.

Answer: $\frac{1}{6}$

3. Find the area of the region bounded above by $y = 4 - x^2$ and below by the x-axis.

$$A = \int_{-2}^2 (4 - x^2) dx$$

- Find where $y = 4 - x^2$ meets the x-axis by solving $4 - x^2 = 0$, giving $x = \pm 2$.
- Set up the integral of $4 - x^2$ from -2 to 2.
- Integrate to get $4x - x^3/3$.
- Evaluate at the bounds: $(8 - 8/3) - (-8 + 8/3) = 32/3$.

Answer: $\frac{32}{3}$

4. Find the area of the region bounded by $y = \sqrt{x}$, the y-axis, and the line $y = 2$.

$$A = \int_0^2 y^2 dy$$

- Sketch the region and note it is easier to integrate with respect to y .
- Rewrite $y = \sqrt{x}$ as $x = y^2$, which is the right boundary.
- Set up the integral of y^2 from $y = 0$ to $y = 2$.
- Integrate to get $y^3/3$ and evaluate to obtain $8/3$.

Answer: $\frac{8}{3}$



5. Find the area of the region bounded by $y = x^3$ and $y = x$ in the first quadrant.

$$A = \int_0^1 (x - x^3) dx$$

→ Find intersections by solving $x = x^3$, giving $x = 0$ and $x = 1$ in the first quadrant.

→ Identify $y = x$ as the top function and $y = x^3$ as the bottom on $[0, 1]$.

→ Integrate $x - x^3$ to get $x^2/2 - x^4/4$.

→ Evaluate from 0 to 1 to get $1/2 - 1/4 = 1/4$.

Answer: $\frac{1}{4}$

6. Find the area of the region bounded by $y = x + 2$, $y = x^2$, and the y-axis using sub-regions if needed.

$$A = \int_0^2 ((x + 2) - x^2) dx$$

→ Find the intersection of $y = x + 2$ and $y = x^2$ by solving $x^2 = x + 2$, giving $x = 2$ (in first quadrant).

→ Determine that $y = x + 2$ is above $y = x^2$ on $[0, 2]$.

→ Set up the integral of $(x + 2) - x^2$ from 0 to 2.

→ Integrate to get $x^2/2 + 2x - x^3/3$ and evaluate to obtain $10/3$.

Answer: $\frac{10}{3}$

7. Find the area of the region bounded by $y = \sqrt{x}$, $y = x - 2$, and the line $x = 0$ using sub-regions.

$$A = \int_0^2 \sqrt{x} dx + \int_2^4 (\sqrt{x} - (x - 2)) dx$$

→ Sketch the curves to see two sub-regions are required.

→ From $x = 0$ to $x = 2$, the bottom is the x-axis since $y = x - 2$ is below it.

→ From $x = 2$ to $x = 4$, the bottom changes to $y = x - 2$.

→ Compute each integral separately and add to get $10/3$.

Answer: $\frac{10}{3}$

8. Find the area enclosed between $y = \sin(x)$ and the x-axis from $x = 0$ to $x = \pi$.

$$A = \int_0^\pi \sin(x) dx$$

→ Note that $\sin(x)$ is non-negative on $[0, \pi]$, so no sub-regions are needed.

→ Set up the definite integral of $\sin(x)$ from 0 to π .

→ The antiderivative of $\sin(x)$ is $-\cos(x)$.

→ Evaluate: $-\cos(\pi) - (-\cos(0)) = 1 + 1 = 2$.

Answer: 2

9. Find the area of the region bounded by $y = x^2$ and $y = 2x$ using top-minus-bottom.

$$A = \int_0^2 (2x - x^2) dx$$

→ Find intersections by solving $x^2 = 2x$, giving $x = 0$ and $x = 2$.

→ Determine $y = 2x$ is above $y = x^2$ on $[0, 2]$.

→ Set up the integral of $2x - x^2$ from 0 to 2.

→ Integrate to get $x^2 - x^3/3$ and evaluate to obtain $4 - 8/3 = 4/3$.

Answer: $\frac{4}{3}$



10. Find the area of the region bounded by $y = \sqrt{x}$, $y = 6 - x$, and the x-axis using sub-regions.

$$A = \int_0^4 \sqrt{x} \, dx + \int_4^6 (6 - x) \, dx$$

→ Find the intersection of $y = \sqrt{x}$ and $y = 6 - x$ by solving $\sqrt{x} = 6 - x$, giving $x = 4$.

→ From $x = 0$ to $x = 4$, the top is $y = \sqrt{x}$ and the bottom is the x-axis.

→ From $x = 4$ to $x = 6$, the top is $y = 6 - x$ and the bottom is the x-axis.

→ Evaluate both integrals: $16/3 + 2 = 22/3$.

Answer: $\frac{22}{3}$

